GATE ECE

Previous Year Questions — Chapter-wise, with Answers and Solutions

Typeset edition

Papers included: GATE 2013.

GATE 2013

GATE 2013 | Q.1 | 1 mark

A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by any one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles

(A) an AND gate
(B) an OR gate
(C) an XOR gate
(D) a NAND gate

Answer: (C) [self-solved, no official key available]

Solution. Toggling either switch toggles the lamp, so the lamp state is $A \oplus B$: an XOR gate.

GATE 2013 | Q.2 | 1 mark

Consider a vector field $\vec{A}(\vec{r})$. The closed loop line integral $\oint \vec{A} \cdot d\vec{l}$ can be expressed as

(A) $\iint (\nabla \times \vec{A}) \cdot d\vec{s}$ over the closed surface bounded by the loop
(B) $\iiint (\nabla \cdot \vec{A})\, dv$ over the closed volume bounded by the loop
(C) $\iiint (\nabla \cdot \vec{A})\, dv$ over the open volume bounded by the loop
(D) $\iint (\nabla \times \vec{A}) \cdot d\vec{s}$ over the open surface bounded by the loop

Answer: (D) [self-solved, no official key available]

Solution. Stokes' theorem: $\oint \vec{A} \cdot d\vec{l} = \iint (\nabla \times \vec{A}) \cdot d\vec{s}$ over any open surface bounded by the loop.

GATE 2013 | Q.3 | 1 mark

Two systems with impulse responses $h_1(t)$ and $h_2(t)$ are connected in cascade. Then the overall impulse response of the cascaded system is given by

(A) product of $h_1(t)$ and $h_2(t)$
(B) sum of $h_1(t)$ and $h_2(t)$
(C) convolution of $h_1(t)$ and $h_2(t)$
(D) subtraction of $h_2(t)$ from $h_1(t)$

Answer: (C) [self-solved, no official key available]

Solution. For a cascade of LTI systems, $h(t) = h_1(t) * h_2(t)$.

GATE 2013 | Q.4 | 1 mark

In a forward biased pn junction diode, the sequence of events that best describes the mechanism of current flow is

(A) injection, and subsequent diffusion and recombination of minority carriers
(B) injection, and subsequent drift and generation of minority carriers

(C) extraction, and subsequent diffusion and generation of minority carriers
(D) extraction, and subsequent drift and recombination of minority carriers

Answer: (A) [self-solved, no official key available]

Solution. Forward bias injects minority carriers across the junction; they diffuse into the neutral region and recombine there.

GATE 2013 | Q.5 | 1 mark

In IC technology, dry oxidation (using dry oxygen) as compared to wet oxidation (using steam or water vapor) produces

(A) superior quality oxide with a higher growth rate
(B) inferior quality oxide with a higher growth rate
(C) inferior quality oxide with a lower growth rate
(D) superior quality oxide with a lower growth rate

Answer: (D) [self-solved, no official key available]

Solution. Dry oxidation grows a denser, higher-quality oxide, but at a lower growth rate than wet oxidation.

GATE 2013 | Q.6 | 2 marks

The maximum value of θ until which the approximation $\sin\theta \approx \theta$ holds to within 10% error is

(A) 10°
(B) 18°
(C) 50°
(D) 90°

Answer: (C) [self-solved, no official key available]

Solution. We need $\sin\theta \geq 0.9\,\theta$. At $\theta = 0.8$ rad ($\approx 46^\circ$), $\sin\theta = 0.717 = 0.9\theta$; hence the limit is about 50°.

GATE 2013 | Q.7 | 2 marks

The divergence of the vector field $\vec{A} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$ is

(A) 0
(B) 1/3
(C) 1
(D) 3

Answer: (D) [self-solved, no official key available]

Solution. $\nabla \cdot \vec{A} = \dfrac{\partial x}{\partial x} + \dfrac{\partial y}{\partial y} + \dfrac{\partial z}{\partial z} = 3.$

GATE 2013 | Q.8 | 2 marks

The impulse response of a system is $h(t) = t\,u(t)$. For an input $u(t-1)$, the output is

(A) $\dfrac{t^2}{2}u(t)$
(B) $\dfrac{t(t-1)}{2}u(t-1)$
(C) $\dfrac{(t-1)^2}{2}u(t-1)$

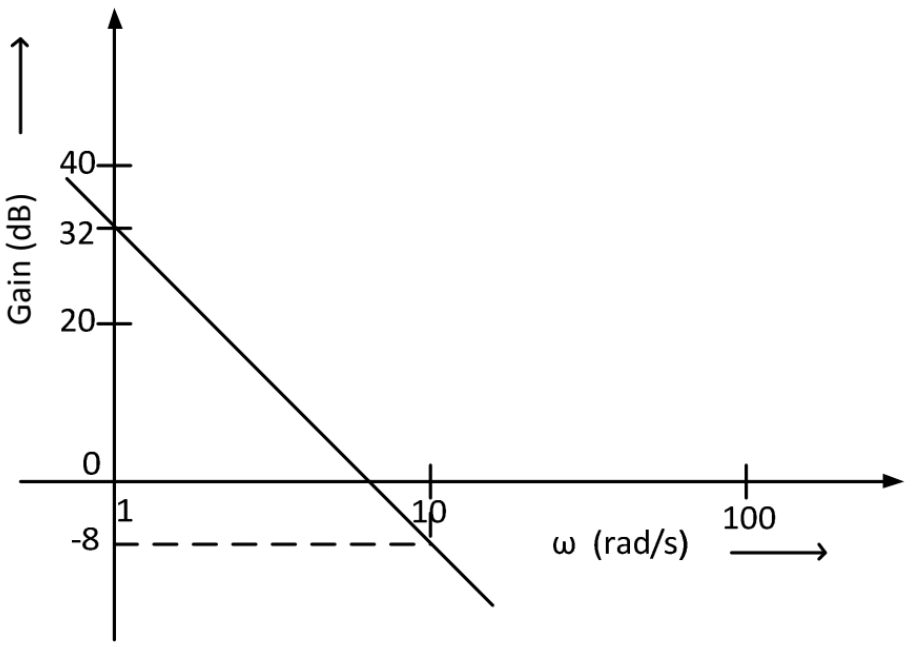
(D) $\dfrac{t^2-1}{2}u(t-1)$

Answer: (C) [self-solved, no official key available]

Solution. $y(t) = h * x$: the ramp $t\,u(t)$ convolved with $u(t-1)$ is the double integral of $\delta(t-1)$, namely $\dfrac{(t-1)^2}{2}u(t-1)$.

GATE 2013 | Q.9 | 2 marks

The Bode plot of a transfer function $G(s)$ is shown in the figure below. The gain $(20\log|G(s)|)$ is 32 dB and -8 dB at 1 rad/s and 10 rad/s respectively. The phase is negative for all ω. Then $G(s)$ is



$\log|G(s)|$) is 32 dB and -8 dB at 1 rad/s and 10 rad/s respective
ω. Then $G(s)$ is

(A) $\dfrac{39.8}{s}$

(B) $\dfrac{39.8}{s^2}$

(C) $\dfrac{32}{s}$

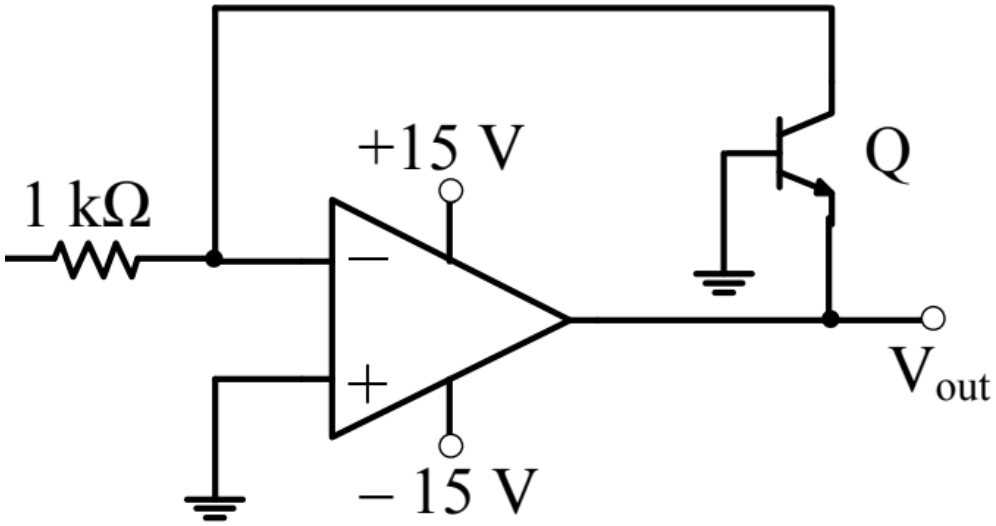
(D) $\dfrac{32}{s^2}$

Answer: (B) [self-solved, no official key available]

Solution. The gain falls by 40 dB per decade (from 32 to -8 dB), so $G = K/s^2$; $20\log K = 32$ gives $K = 10^{1.6} = 39.8$.

GATE 2013 | Q.10 | 2 marks

In the circuit shown below what is the output voltage (V_{out}) if a silicon transistor Q and an ideal op-amp are used?



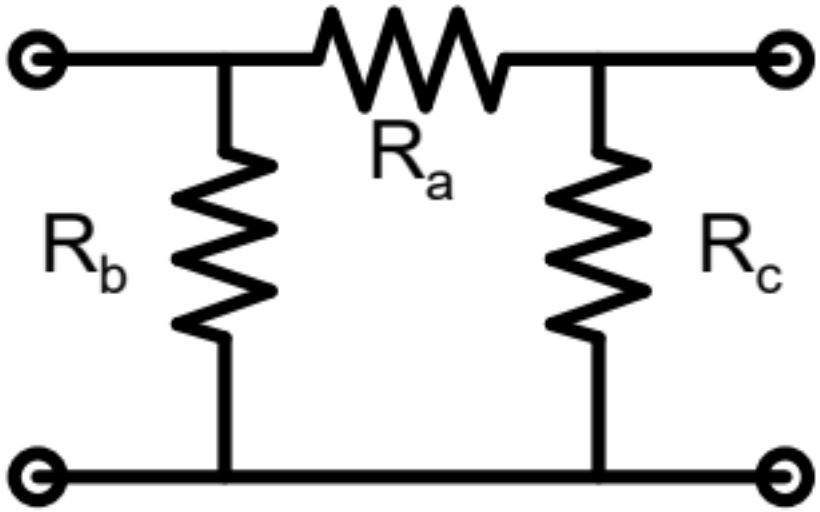
(A) −15 V
(B) −0.7 V
(C) +0.7 V
(D) +15 V

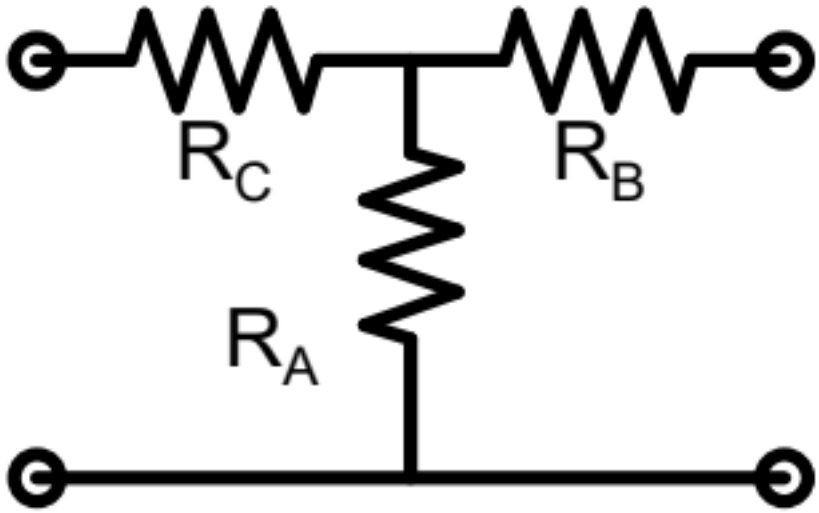
Answer: (B) [self-solved, no official key available]

Solution. The transistor in the feedback path conducts so that the inverting node stays at 0 V; V_{out} is one base–emitter drop below that: −0.7 V.

GATE 2013 | Q.11 | 1 mark

Consider a delta connection of resistors and its equivalent star connection as shown below. If all elements of the delta connection are scaled by a factor k, $k > 0$, the elements of the corresponding star equivalent will be scaled by a factor of





(A) k^2
(B) k
(C) $1/k$
(D) $\sqrt{k}$

Answer: (B) [self-solved, no official key available]

Solution. $R_a = \dfrac{R_b R_c}{R_a + R_b + R_c}$ is homogeneous of degree one in the resistances, so each star resistance scales by k.

GATE 2013 | Q.12 | 1 mark

For 8085 microprocessor, the following program is executed.
`MVI A, 05H; MVI B, 05H; PTR: ADD B; DCR B; JNZ PTR; ADI 03H; HLT.`
At the end of program, accumulator contains

(A) 17H
(B) 20H
(C) 23H
(D) 05H

Answer: (A) [self-solved, no official key available]

Solution. The loop gives $A = 5+5+4+3+2+1 = 20 = 14\text{H}$ when B becomes 0; then `ADI 03H` gives 17H.

GATE 2013 | Q.13 | 1 mark

The bit rate of a digital communication system is R kbits/s. The modulation used is 32-QAM. The minimum bandwidth required for ISI free transmission is

(A) $R/10$ Hz
(B) $R/10$ kHz
(C) $R/5$ Hz
(D) $R/5$ kHz

Answer: (B) [self-solved, no official key available]

Solution. 32-QAM carries 5 bits/symbol, so the symbol rate is $R/5$ kbaud; the minimum (Nyquist) bandwidth is half of it: $R/10$ kHz.

GATE 2013 | Q.14 | 1 mark

For a periodic signal $v(t) = 30\sin 100t + 10\cos 300t + 6\sin\left(500t + \frac{\pi}{4}\right)$, the fundamental frequency in rad/s is

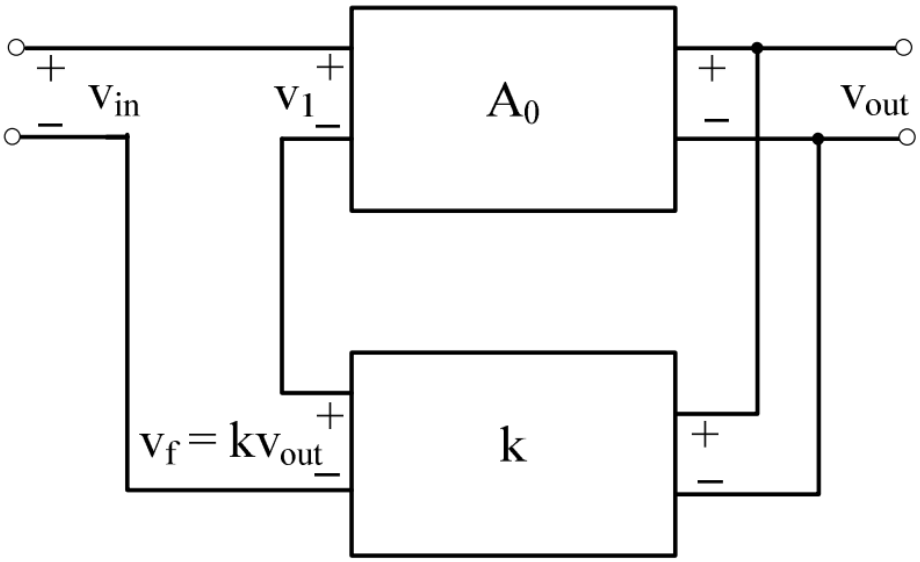
(A) 100
(B) 300
(C) 500
(D) 1500

Answer: (A) [self-solved, no official key available]

Solution. The components are at 100, 300, 500 rad/s; the fundamental is their greatest common divisor, 100 rad/s.

GATE 2013 | Q.15 | 1 mark

In a voltage-voltage feedback as shown below, which one of the following statements is TRUE if the gain k is increased?



(A) The input impedance increases and output impedance decreases.
(B) The input impedance increases and output impedance also increases.
(C) The input impedance decreases and output impedance also decreases.
(D) The input impedance decreases and output impedance increases.

Answer: (A) [self-solved, no official key available]

Solution. Series–shunt (voltage–voltage) feedback multiplies the input impedance and divides the output impedance by $(1 + A_0 k)$; both effects grow with k.

GATE 2013 | Q.16 | 1 mark

A band-limited signal with a maximum frequency of 5 kHz is to be sampled. According to the sampling theorem, the sampling frequency which is not valid is

(A) 5 kHz

(B) 12 kHz
(C) 15 kHz
(D) 20 kHz

Answer: (A) [self-solved, no official key available]

Solution. The sampling frequency must exceed $2f_m = 10$ kHz, so 5 kHz is not valid.

GATE 2013 | Q.17 | 1 mark

In a MOSFET operating in the saturation region, the channel length modulation effect causes

(A) an increase in the gate-source capacitance
(B) a decrease in the transconductance
(C) a decrease in the unity-gain cutoff frequency
(D) a decrease in the output resistance

Answer: (D) [self-solved, no official key available]

Solution. Channel-length modulation makes I_D depend on V_{DS}, i.e. it introduces a finite output conductance and so reduces the output resistance r_o.

GATE 2013 | Q.18 | 1 mark

Which one of the following statements is NOT TRUE for a continuous time causal and stable LTI system?

(A) All the poles of the system must lie on the left side of the $j\omega$ axis.
(B) Zeros of the system can lie anywhere in the s-plane.
(C) All the poles must lie within $|s| = 1$.
(D) All the roots of the characteristic equation must be located on the left side of the $j\omega$ axis.

Answer: (C) [self-solved, no official key available]

Solution. Causality and stability only require the poles to lie in the open left half-plane; they need not lie inside the unit circle.

GATE 2013 | Q.19 | 1 mark

The minimum eigenvalue of the following matrix is

$$\begin{bmatrix} 3 & 5 & 2 \\ 5 & 12 & 7 \\ 2 & 7 & 5 \end{bmatrix}$$

(A) 0
(B) 1
(C) 2
(D) 3

Answer: (A) [self-solved, no official key available]

Solution. Row 2 equals Row 1 plus Row 3, so the matrix is singular and 0 is an eigenvalue; the symmetric matrix is positive semidefinite, so 0 is the minimum.

GATE 2013 | Q.20 | 1 mark

A polynomial $f(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x - a_0$ with all coefficients positive has

(A) no real roots

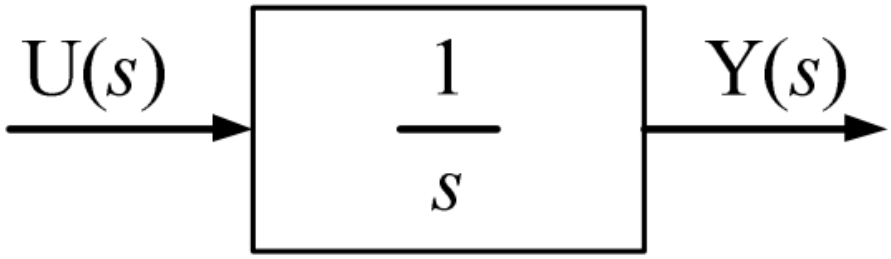
(B) no negative real root
(C) odd number of real roots
(D) at least one positive and one negative real root

Answer: (D) [self-solved, no official key available]

Solution. $f(0) = -a_0 < 0$ while $f(x) \to +\infty$ as $x \to \pm\infty$ (even degree, positive leading coefficient), so there is a root on each side of 0.

GATE 2013 | Q.21 | 1 mark

Assuming zero initial condition, the response $y(t)$ of the system given below to a unit step input $u(t)$ is



$t\,u(t)$ (C) $\frac{t^2}{2}u(t)$

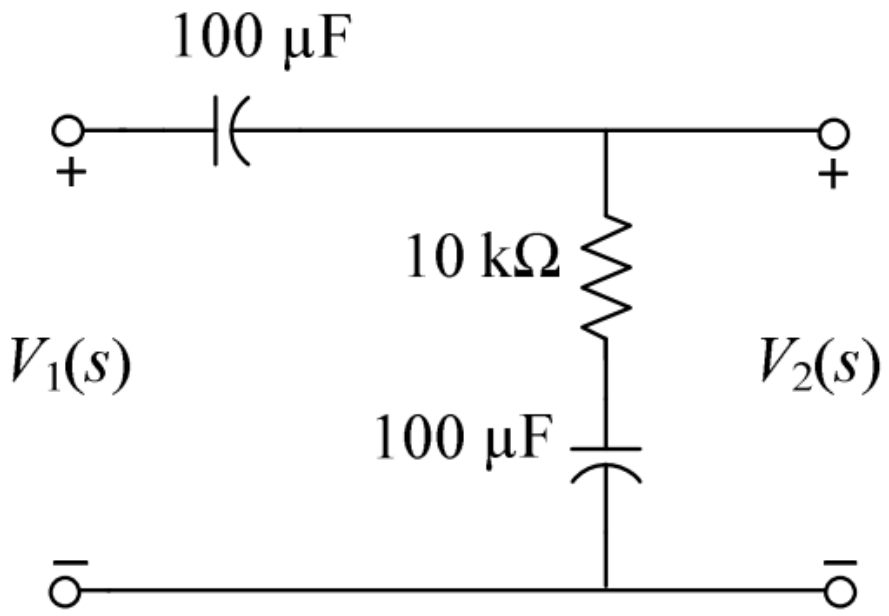
(A) $u(t)$
(B) $t\,u(t)$
(C) $\frac{t^2}{2}u(t)$
(D) $e^{-t}u(t)$

Answer: (B) [self-solved, no official key available]

Solution. The block $1/s$ integrates its input: $\int_0^t u = t\,u(t)$.

GATE 2013 | Q.22 | 1 mark

The transfer function $\frac{V_2(s)}{V_1(s)}$ of the circuit shown below is



(A) $\dfrac{0.5s+1}{s+1}$
(B) $\dfrac{3s+6}{s+2}$
(C) $\dfrac{s+2}{s+1}$
(D) $\dfrac{s+1}{s+2}$

Answer: (D) [self-solved, no official key available]

Solution. The series capacitor has $Z = 10^4/s\ \Omega$ and the shunt branch $Z = 10^4 + 10^4/s = 10^4(s+1)/s$. Therefore $\dfrac{V_2}{V_1} = \dfrac{(s+1)/s}{(s+1)/s + 1/s} = \dfrac{s+1}{s+2}$.

GATE 2013 | Q.23 | 1 mark

A source $v_s(t) = V\cos 100\pi t$ has an internal impedance of $(4+j3)\ \Omega$. If a purely resistive load connected to this source has to extract the maximum power out of the source, its value in Ω should be

(A) 3
(B) 4
(C) 5
(D) 7

Answer: (C) [self-solved, no official key available]

Solution. For a purely resistive load the maximum power is delivered when $R_L = |Z_s| = \sqrt{4^2+3^2} = 5\ \Omega$.

GATE 2013 | Q.24 | 1 mark

The return loss of a device is found to be 20 dB. The voltage standing wave ratio (VSWR) and magnitude of reflection coefficient are respectively

(A) 1.22 and 0.1
(B) 0.81 and 0.1

(C) -1.22 and 0.1
(D) 2.44 and 0.2

Answer: (A) [self-solved, no official key available]

Solution. $RL = -20\log|\Gamma| = 20 \Rightarrow |\Gamma| = 0.1$; VSWR $= \dfrac{1+0.1}{1-0.1} = 1.22$.

GATE 2013 | Q.25 | 1 mark

Let $g(t) = e^{-\pi t^2}$, and $h(t)$ is a filter matched to $g(t)$. If $g(t)$ is applied as input to $h(t)$, then the Fourier transform of the output is

(A) $e^{-\pi f^2}$
(B) $e^{-\pi f^2/2}$
(C) $e^{-\pi|f|}$
(D) $e^{-2\pi f^2}$

Answer: (D) [self-solved, no official key available]

Solution. The matched-filter output is the autocorrelation of g, whose transform is $|G(f)|^2 = \left(e^{-\pi f^2}\right)^2 = e^{-2\pi f^2}$.

GATE 2013 | Q.26 | 1 mark

Let U and V be two independent zero mean Gaussian random variables of variances $\dfrac{1}{4}$ and $\dfrac{1}{9}$ respectively. The probability $P(3V \geq 2U)$ is

(A) 4/9
(B) 1/2
(C) 2/3
(D) 5/9

Answer: (B) [self-solved, no official key available]

Solution. $3V - 2U$ is a zero-mean Gaussian variable (variance $9 \cdot \frac{1}{9} + 4 \cdot \frac{1}{4} = 2$), symmetric about 0, so $P(3V - 2U \geq 0) = \frac{1}{2}$.

GATE 2013 | Q.27 | 1 mark

Let A be an $m \times n$ matrix and B an $n \times m$ matrix. It is given that $\det(I_m + AB) = \det(I_n + BA)$, where I_k is the $k \times k$ identity matrix. Using the above property, the determinant of the matrix given below is

$$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

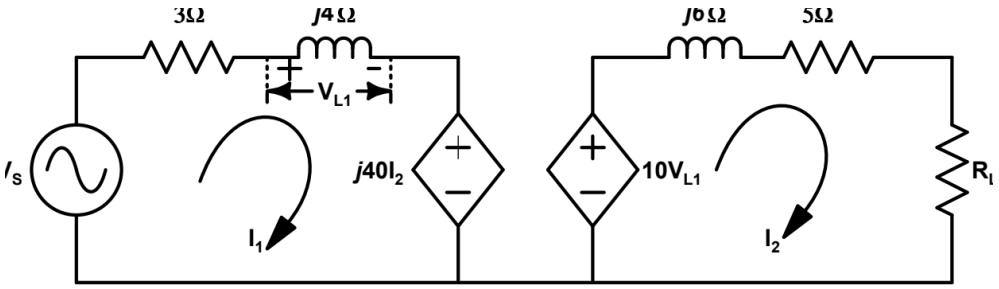
(A) 2
(B) 5
(C) 8
(D) 16

Answer: (B) [self-solved, no official key available]

Solution. The matrix is $I_4 + uu^T$ with $u = (1,1,1,1)^T$. Then $\det = \det(1 + u^T u) = 1 + 4 = 5$.

GATE 2013 | Q.28 | 1 mark

In the circuit shown below, if the source voltage $V_S = 100\angle 53.13°$ V then the Thevenin's equivalent voltage in Volts as seen by the load resistance R_L is



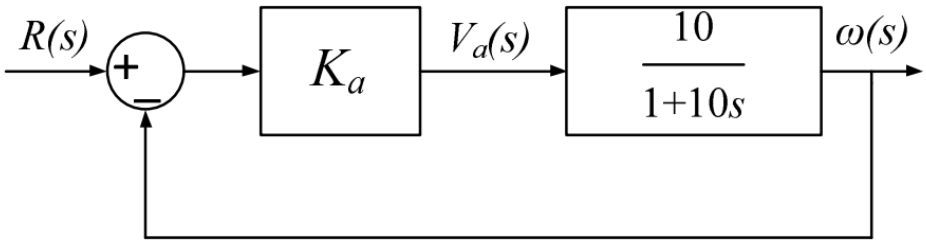
(A) $100\angle 90°$
(B) $800\angle 0°$
(C) $800\angle 90°$
(D) $100\angle 60°$

Answer: (C) [self-solved, no official key available]

Solution. With the load removed, $I_2 = 0$ and $I_1 = \dfrac{V_S}{3+j4} = 20\angle 0°$ A; $V_{L1} = j4I_1 = 80\angle 90°$ V. The dependent source gives $V_{Th} = 10V_{L1} = 800\angle 90°$ V.

GATE 2013 | Q.29 | 1 mark

The open-loop transfer function of a dc motor is given as $\dfrac{\omega(s)}{V_a(s)} = \dfrac{10}{1+10s}$. When connected in feedback as shown below, the approximate value of K_a that will reduce the time constant of the closed loop system by one hundred times as compared to that of the open-loop system is



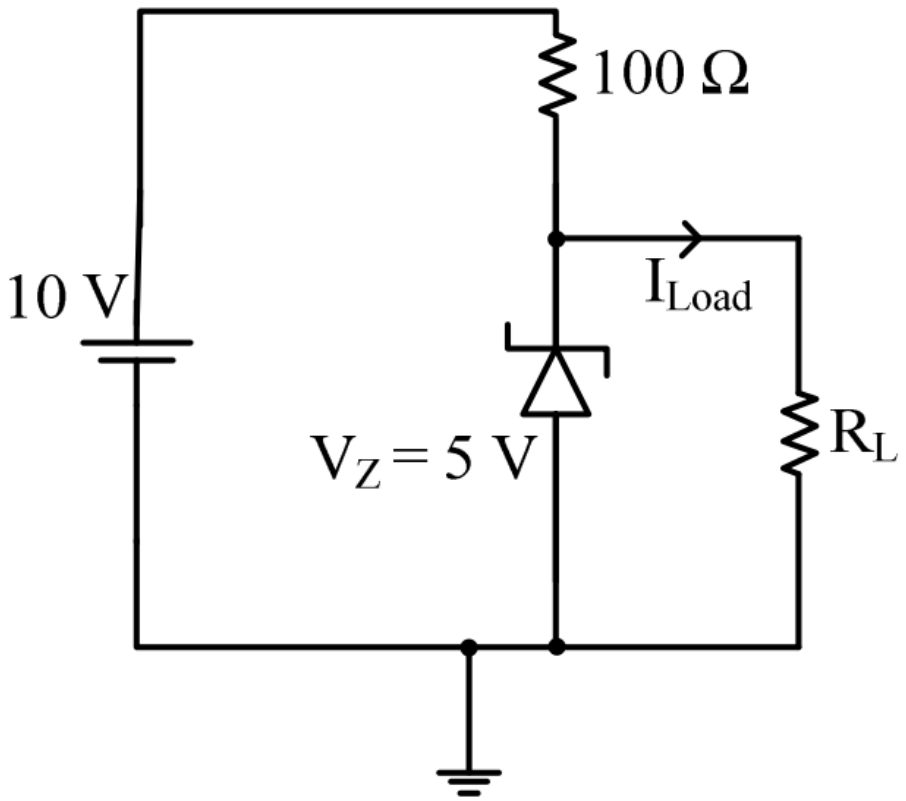
(A) 1
(B) 5
(C) 10
(D) 100

Answer: (C) [self-solved, no official key available]

Solution. The closed-loop pole is at $s = -\dfrac{1+10K_a}{10}$, so $\tau_{cl} = \dfrac{10}{1+10K_a}$. For $\tau_{cl} = \dfrac{10}{100}$ we need $1+10K_a = 100$, i.e. $K_a \approx 10$.

GATE 2013 | Q.30 | 1 mark

In the circuit shown below, the knee current of the ideal Zener diode is 10 mA. To maintain 5 V across R_L, the minimum value of R_L in Ω and the minimum power rating of the Zener diode in mW, respectively, are



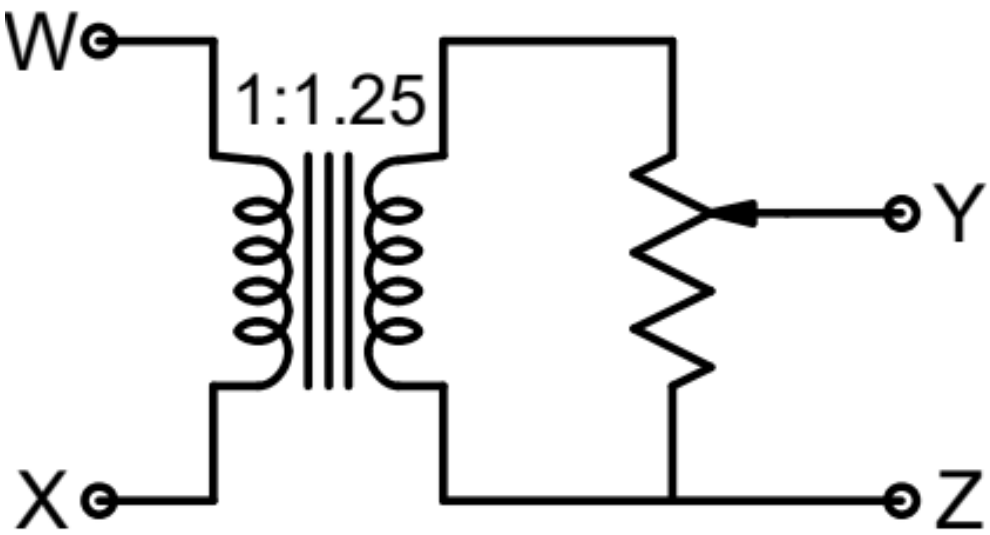
(A) 125 and 125
(B) 125 and 250
(C) 250 and 125
(D) 250 and 250

Answer: (B) [self-solved, no official key available]

Solution. The series resistor carries $(10-5)/100 = 50$ mA. Since the Zener needs at least 10 mA, $I_L \leq 40$ mA, so $R_{L,\min} = 5/0.04 = 125\ \Omega$. The Zener dissipates the most when $I_L = 0$: 5×50 mA $= 250$ mW.

GATE 2013 | Q.31 | 1 mark

The following arrangement consists of an ideal transformer and an attenuator which attenuates by a factor of 0.8. An ac voltage $V_{WX1} = 100$ V is applied across WX to get an open circuit voltage V_{YZ1} across YZ. Next, an ac voltage $V_{YZ2} = 100$ V is applied across YZ to get an open circuit voltage V_{WX2} across WX. Then, V_{YZ1}/V_{WX1}, V_{WX2}/V_{YZ2} are respectively,



(A) 125/100 and 80/100
(B) 100/100 and 80/100
(C) 100/100 and 100/100
(D) 80/100 and 80/100

Answer: (C) [self-solved, no official key available]

Solution. Forward: $1.25 \times 0.8 = 1$, so $V_{YZ1}/V_{WX1} = 100/100$. The arrangement is reciprocal, and the ratio is the same in the reverse direction: $V_{WX2}/V_{YZ2} = 100/100$.

GATE 2013 | Q.32 | 1 mark

Two magnetically uncoupled inductive coils have Q factors q_1 and q_2 at the chosen operating frequency. Their respective resistances are R_1 and R_2. When connected in series, their effective Q factor at the same operating frequency is

(A) $q_1 + q_2$
(B) $(1/q_1) + (1/q_2)$
(C) $(q_1R_1 + q_2R_2)/(R_1 + R_2)$
(D) $(q_1R_2 + q_2R_1)/(R_1 + R_2)$

Answer: (C) [self-solved, no official key available]

Solution. $Q = \dfrac{\omega(L_1 + L_2)}{R_1 + R_2} = \dfrac{q_1R_1 + q_2R_2}{R_1 + R_2}$ since $\omega L_i = q_iR_i$.

GATE 2013 | Q.33 | 1 mark

The impulse response of a continuous time system is given by $h(t) = \delta(t-1) + \delta(t-3)$. The value of the step response at $t = 2$ is

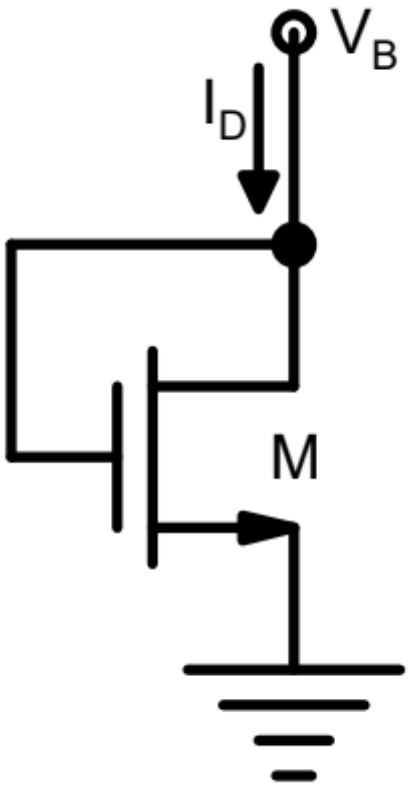
(A) 0
(B) 1
(C) 2
(D) 3

Answer: (B) [self-solved, no official key available]

Solution. $s(t) = u(t-1) + u(t-3)$, so $s(2) = 1 + 0 = 1$.

GATE 2013 | Q.34 | 1 mark

The small-signal resistance (i.e., dV_B/dI_D) in kΩ offered by the n-channel MOSFET M shown in the figure below, at a bias point of $V_B = 2$ V is (device data for M: device transconductance parameter $k_N = \mu_n C'_{ox}(W/L) = 40\ \mu\text{A/V}^2$, threshold voltage $V_{TN} = 1$ V, and neglect body effect and channel length modulation effects)



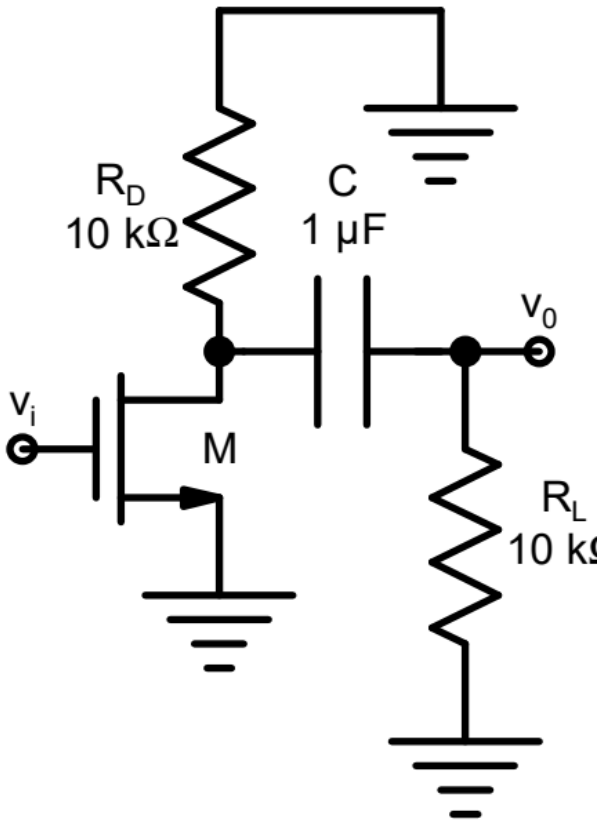
(A) 12.5
(B) 25
(C) 50
(D) 100

Answer: (B) [self-solved, no official key available]

Solution. The gate is tied to the drain, so $I_D = \frac{k_N}{2}(V_B - V_T)^2 = 20\ \mu\text{A}$ at $V_B = 2$ V and $\frac{dI_D}{dV_B} = k_N(V_B - V_T) = 40\ \mu\text{S}$, giving $r = 25$ kΩ.

GATE 2013 | Q.35 | 1 mark

The ac schematic of an NMOS common-source stage is shown in the figure below, where part of the biasing circuits has been omitted for simplicity. For the n-channel MOSFET M, the transconductance $g_m = 1$ mA/V, and body effect and channel length modulation effect are to be neglected. The lower cutoff frequency in Hz of the circuit is approximately at



(A) 8
(B) 32
(C) 50
(D) 200

Answer: (A) [self-solved, no official key available]

Solution. The coupling capacitor sees $R_D + R_L = 20$ kΩ: $f_L = \dfrac{1}{2\pi(R_D + R_L)C} = \dfrac{1}{2\pi \cdot 20\text{k} \cdot 1\mu} \approx 8$ Hz.

GATE 2013 | Q.36 | 2 marks

A system is described by the differential equation $\dfrac{d^2y}{dt^2} + 5\dfrac{dy}{dt} + 6y(t) = x(t)$. Let $x(t)$ be a rectangular pulse given by $x(t) = \begin{cases} 1 & 0 < t < 2 \\ 0 & \text{otherwise} \end{cases}$. Assuming that $y(0) = 0$ and $\dfrac{dy}{dt} = 0$ at $t = 0$, the Laplace transform of $y(t)$ is

(A) $\dfrac{e^{-2s}}{s(s+2)(s+3)}$
(B) $\dfrac{1 - e^{-2s}}{s(s+2)(s+3)}$
(C) $\dfrac{e^{-2s}}{(s+2)(s+3)}$
(D) $\dfrac{1 - e^{-2s}}{(s+2)(s+3)}$

Answer: (B) [self-solved, no official key available]

Solution. With zero initial conditions, $Y(s) = \dfrac{X(s)}{s^2+5s+6}$ and $X(s) = \dfrac{1-e^{-2s}}{s}$.

GATE 2013 | Q.37 | 2 marks

A system described by a linear, constant coefficient, ordinary, first order differential equation has an exact solution given by $y(t)$ for $t > 0$, when the forcing function is $x(t)$ and the initial condition is $y(0)$. If one wishes to modify the system so that the solution becomes $-2y(t)$ for $t > 0$, we need to

(A) change the initial condition to $-y(0)$ and the forcing function to $2x(t)$
(B) change the initial condition to $2y(0)$ and the forcing function to $-x(t)$
(C) change the initial condition to $j\sqrt{2}\,y(0)$ and the forcing function to $j\sqrt{2}\,x(t)$
(D) change the initial condition to $-2y(0)$ and the forcing function to $-2x(t)$

Answer: (D) [self-solved, no official key available]

Solution. The response is linear in the pair $(y(0), x)$: $y = y_{zi} + y_{zs}$. Scaling both the initial condition and the forcing function by -2 scales the solution by -2.

GATE 2013 | Q.38 | 2 marks

Consider two identically distributed zero-mean random variables U and V. Let the cumulative distribution functions of U and $2V$ be $F(x)$ and $G(x)$ respectively. Then, for all values of x

(A) $F(x) - G(x) \le 0$
(B) $F(x) - G(x) \ge 0$
(C) $(F(x) - G(x)) \cdot x \le 0$
(D) $(F(x) - G(x)) \cdot x \ge 0$

Answer: (D) [self-solved, no official key available]

Solution. $G(x) = P(2V \le x) = F(x/2)$. For $x \ge 0$, $F(x) \ge F(x/2)$; for $x < 0$, $F(x) \le F(x/2)$. Hence $(F - G)\,x \ge 0$.

GATE 2013 | Q.39 | 2 marks

The DFT of a vector $[a\ b\ c\ d]$ is the vector $[\alpha\ \beta\ \gamma\ \delta]$. Consider the product

$$[p\ q\ r\ s] = [a\ b\ c\ d] \begin{bmatrix} a & b & c & d \\ d & a & b & c \\ c & d & a & b \\ b & c & d & a \end{bmatrix}.$$

The DFT of the vector $[p\ q\ r\ s]$ is a scaled version of

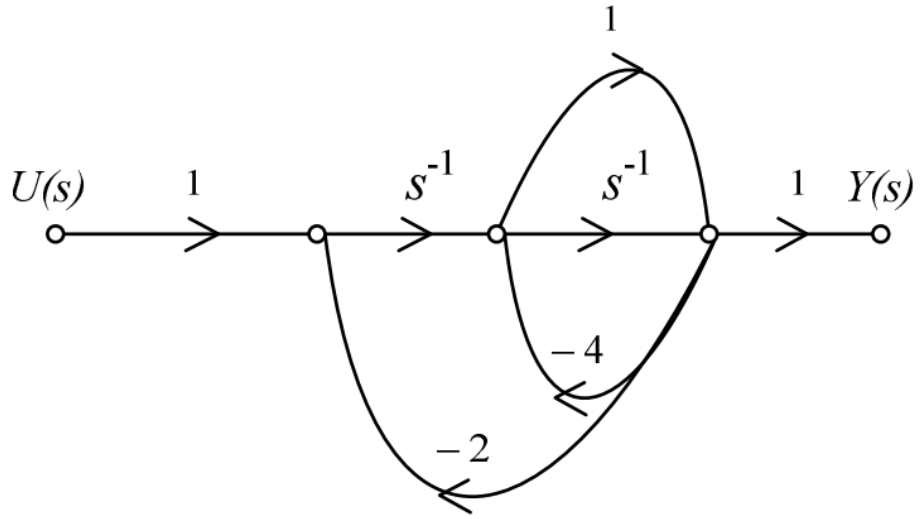
(A) $[\alpha^2\ \beta^2\ \gamma^2\ \delta^2]$
(B) $[\sqrt{\alpha}\ \sqrt{\beta}\ \sqrt{\gamma}\ \sqrt{\delta}]$
(C) $[\alpha+\beta\ \ \beta+\delta\ \ \delta+\gamma\ \ \gamma+\alpha]$
(D) $[\alpha\ \beta\ \gamma\ \delta]$

Answer: (A) [self-solved, no official key available]

Solution. The matrix is circulant, so the product is the circular convolution of $[a\,b\,c\,d]$ with itself. Convolution corresponds to multiplication of DFTs, giving $[\alpha^2\ \beta^2\ \gamma^2\ \delta^2]$.

GATE 2013 | Q.40 | 2 marks

The signal flow graph for a system is given below. The transfer function $\dfrac{Y(s)}{U(s)}$ for this system is



(B) $\dfrac{s+1}{s^2+6s+2}$

(A) $\dfrac{s+1}{5s^2+6s+2}$

(B) $\dfrac{s+1}{s^2+6s+2}$

(C) $\dfrac{s+1}{s^2+4s+2}$

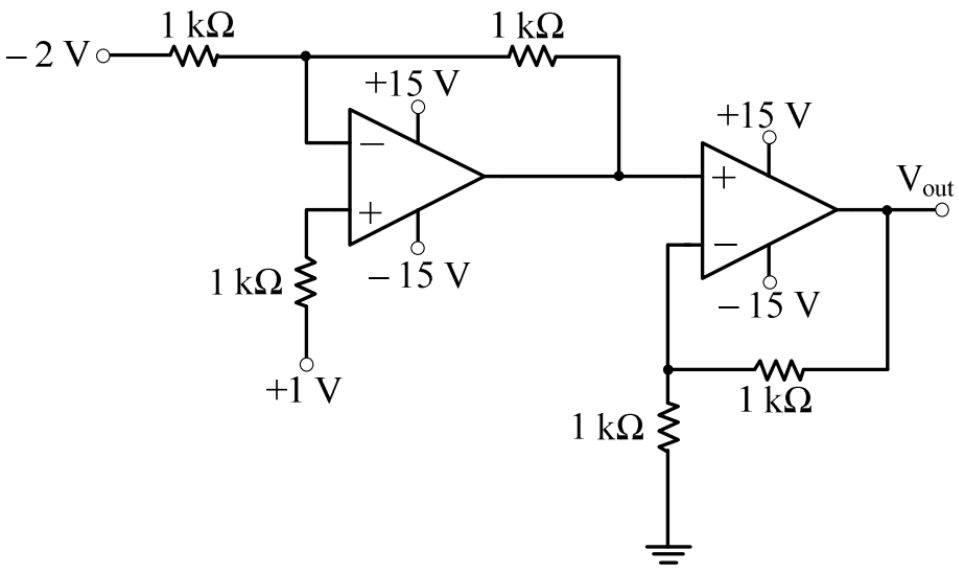
(D) $\dfrac{1}{5s^2+6s+2}$

Answer: (A) [self-solved, no official key available]

Solution. Forward paths: s^{-2} and s^{-1}. Loops: $-4s^{-1}$, -4, $-2s^{-2}$, $-2s^{-1}$ (all touching). $\Delta = 1 + 6s^{-1} + 4 + 2s^{-2}$, so $\dfrac{Y}{U} = \dfrac{s^{-2}+s^{-1}}{5+6s^{-1}+2s^{-2}} = \dfrac{s+1}{5s^2+6s+2}$.

GATE 2013 | Q.41 | 2 marks

In the circuit shown below the op-amps are ideal. Then V_{out} in Volts is



(A) 4
(B) 6
(C) 8
(D) 10

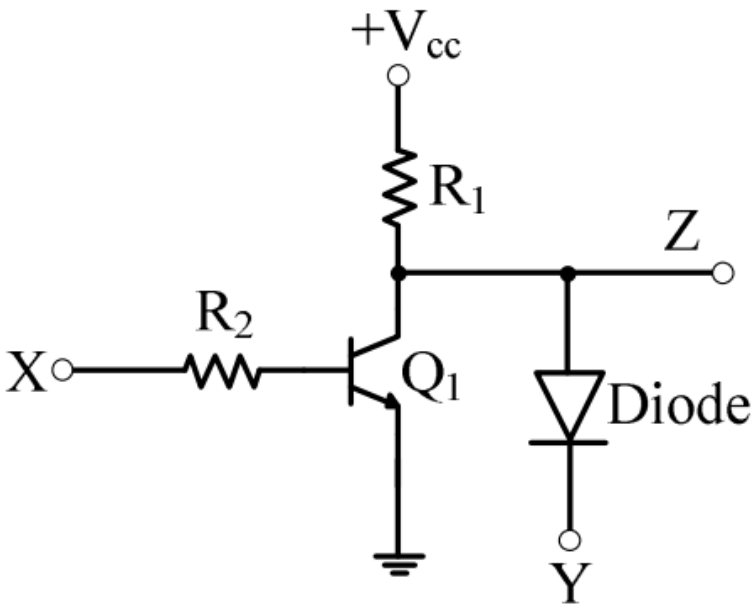
Answer: (C) [self-solved, no official key available]

Solution. Stage 1: $V_{o1} = \left(1 + \frac{1}{1}\right)(1) - \frac{1}{1}(-2) = 4$ V. Stage 2 is non-inverting with gain 2: $V_{out} = 8$ V.

GATE 2013 | Q.42 | 2 marks

In the circuit shown below, Q_1 has negligible collector-to-emitter saturation voltage and the diode drops negligible voltage across it under forward bias. If V_{cc} is +5 V, X and Y are digital signals with 0 V as logic 0 and V_{cc} as logic 1, then the Boolean expression for Z is

$_{CC}$ as logic 1, then the Boolean expression fo



) $\overline{X}\,Y$ (C) $X\,\overline{Y}$

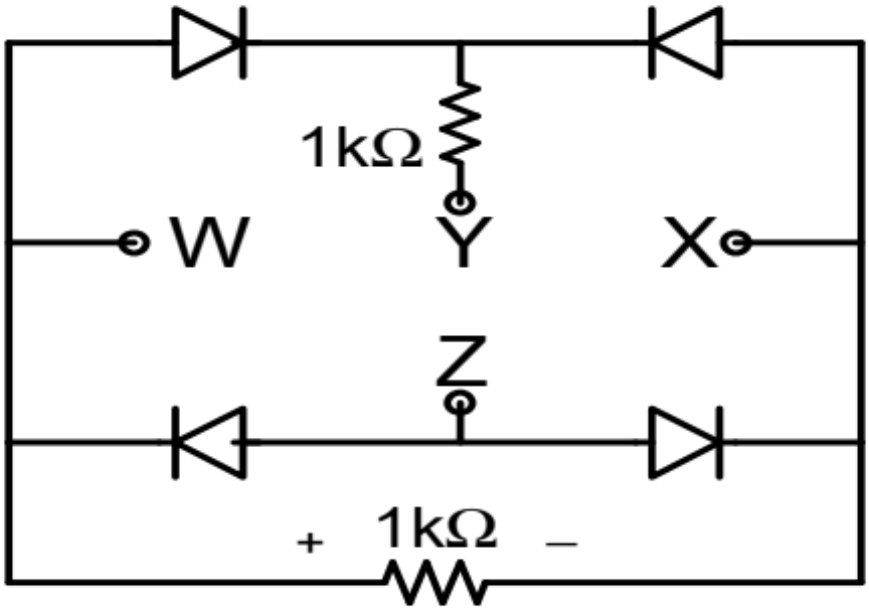
(A) XY
(B) $\overline{X}Y$
(C) $X\overline{Y}$
(D) $\overline{XY}$

Answer: (B) [self-solved, no official key available]

Solution. $Z = 0$ if Q_1 is on ($X = 1$) or if the diode conducts to ground ($Y = 0$). $Z = 1$ only for $X = 0$, $Y = 1$: $Z = \overline{X}Y$.

GATE 2013 | Q.43 | 2 marks

A voltage $1000 \sin \omega t$ Volts is applied across YZ. Assuming ideal diodes, the voltage measured across WX in Volts, is



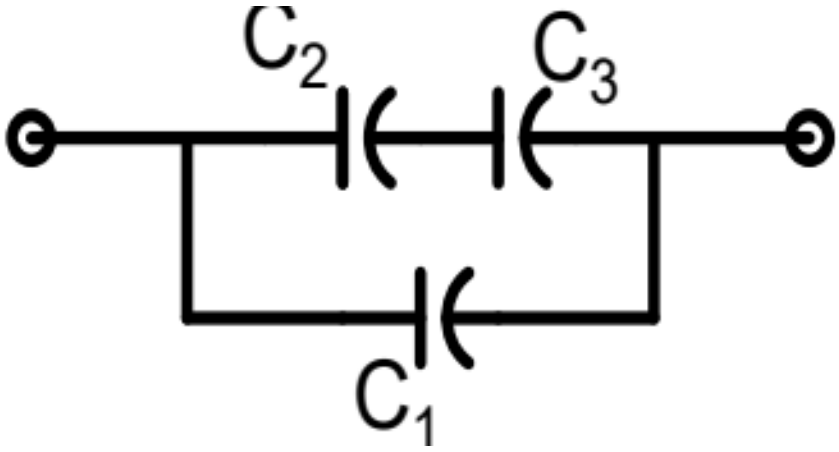
(A) $\sin \omega t$
(B) $(\sin \omega t + |\sin \omega t|)/2$
(C) $(\sin \omega t - |\sin \omega t|)/2$
(D) 0 for all t

Answer: (D) [self-solved, no official key available]

Solution. The diodes block conduction between W and X for either polarity of V_{YZ}, so the measured voltage across WX is 0 at all times.

GATE 2013 | Q.44 | 2 marks

Three capacitors C_1, C_2 and C_3 whose values are 10 μF, 5 μF, and 2 μF respectively, have breakdown voltages of 10 V, 5 V, and 2 V respectively. For the interconnection shown below, the maximum safe voltage in Volts that can be applied across the combination, and the corresponding total charge in μC stored in the effective capacitance across the terminals are respectively,



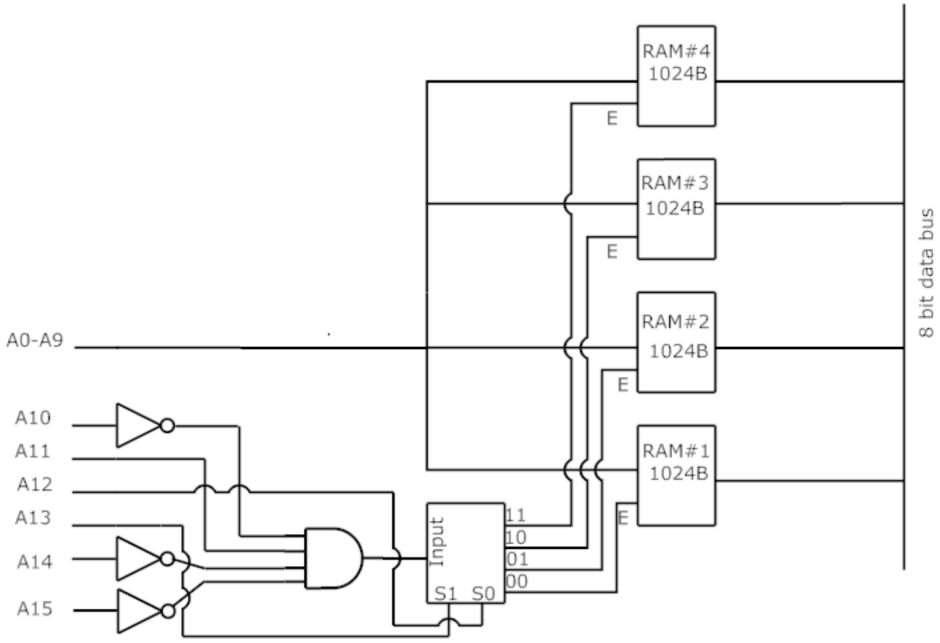
(A) 2.8 and 36
(B) 7 and 119
(C) 2.8 and 32
(D) 7 and 80

Answer: (C) [self-solved, no official key available]

Solution. C_2 and C_3 in series carry the same charge; C_3 limits it to $2 \times 2 = 4$ μC, i.e. $V = 4/(10/7) = 2.8$ V for the series branch (and for C_1 in parallel). $C_{eq} = 10 + \frac{10}{7} = 11.43$ μF, so $Q = 11.43 \times 2.8 = 32$ μC.

GATE 2013 | Q.45 | 2 marks

There are four chips each of 1024 bytes connected to a 16 bit address bus as shown in the figure below. RAMs 1, 2, 3 and 4 respectively are mapped to addresses



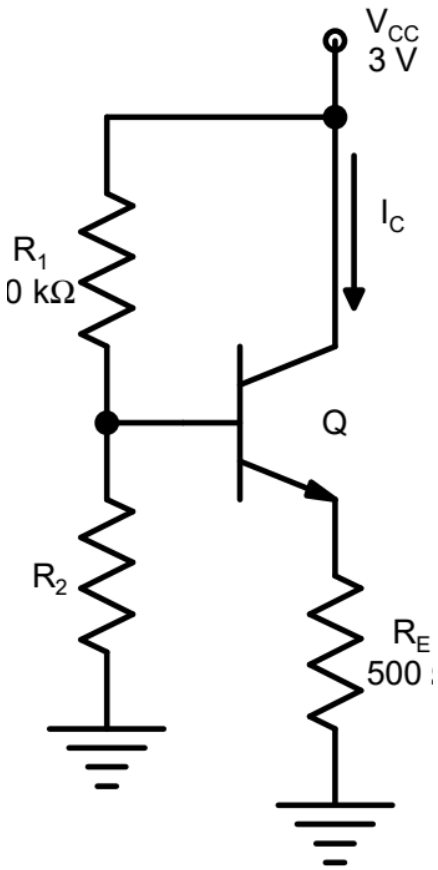
(A) 0C00H-0FFFH, 1C00H-1FFFH, 2C00H-2FFFH, 3C00H-3FFFH
(B) 1800H-1FFFH, 2800H-2FFFH, 3800H-3FFFH, 4800H-4FFFH
(C) 0500H-08FFH, 1500H-18FFH, 2500H-28FFH, 3500H-38FFH
(D) 0800H-0BFFH, 1800H-1BFFH, 2800H-2BFFH, 3800H-3BFFH

Answer: (A) [self-solved, no official key available]

Solution. Each chip occupies 400H addresses (A_0–A_9); with the decoder inputs A_{12}, A_{13} and the enable formed from A_{10}, A_{11}, A_{14}, A_{15} as drawn, the chips are at 0C00H–0FFFH, 1C00H–1FFFH, ...

GATE 2013 | Q.46 | 2 marks

In the circuit shown below, the silicon *npn* transistor Q has a very high value of β. The required value of R_2 in kΩ to produce $I_C = 1$ mA is



(A) 20
(B) 30
(C) 40
(D) 50

Answer: (C) [self-solved, no official key available]

Solution. $V_E = 1\text{ mA} \times 500\ \Omega = 0.5$ V, $V_B = 1.2$ V. The current through R_1 is $\frac{3-1.2}{60\text{k}} = 30\ \mu$A (base current negligible), so $R_2 = \dfrac{1.2}{30\ \mu\text{A}} = 40\ \text{k}\Omega$.

GATE 2013 | Q.47 | 2 marks

Let U and V be two independent and identically distributed random variables such that $P(U = +1) = P(U = -1) = \dfrac{1}{2}$. The entropy $H(U + V)$ in bits is

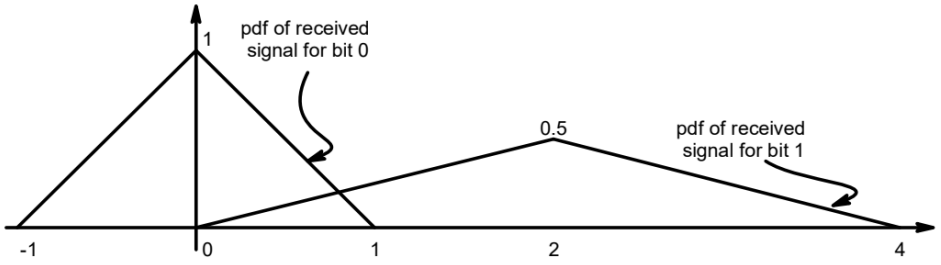
(A) 3/4
(B) 1
(C) 3/2
(D) $\log_2 3$

Answer: (C) [self-solved, no official key available]

Solution. $U + V \in \{-2, 0, 2\}$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$: $H = \frac{1}{4} \cdot 2 + \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 = 1.5$ bits.

GATE 2013 | Q.48 | 2 marks

Common Data for Questions 48 and 49:
Bits 1 and 0 are transmitted with equal probability. At the receiver, the pdf of the respective received signals for both bits are as shown below.
Q.48 If the detection threshold is 1, the BER will be



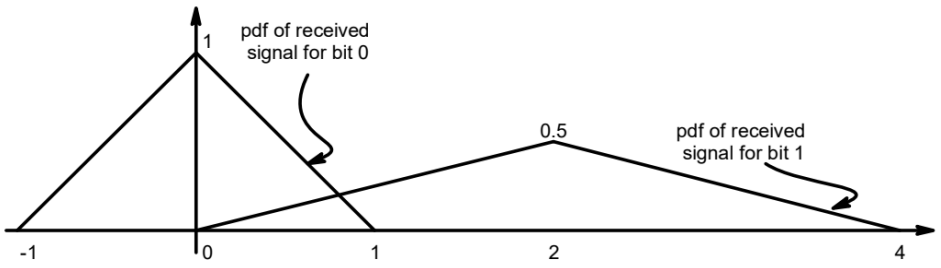
(A) 1/2
(B) 1/4
(C) 1/8
(D) 1/16

Answer: (D) [self-solved, no official key available]

Solution. For threshold 1: bit 0 is never decided as 1 (its pdf is zero above 1). For bit 1 the pdf rises linearly from 0 at $x = 0$ to 0.25 at $x = 1$: error probability $= \frac{1}{2} \cdot 1 \cdot 0.25 = \frac{1}{8}$. BER $= \frac{1}{2}\left(0 + \frac{1}{8}\right) = \frac{1}{16}$.

GATE 2013 | Q.49 | 2 marks

Common Data for Questions 48 and 49 (figure above).
Q.49 The optimum threshold to achieve minimum bit error rate (BER) is



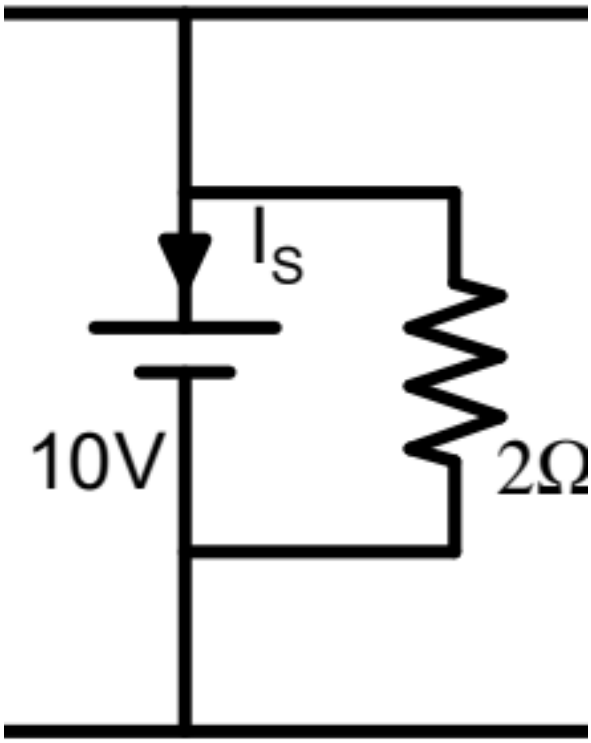
(A) 1/2
(B) 4/5
(C) 1
(D) 3/2

Answer: (B) [self-solved, no official key available]

Solution. The optimum threshold is where the two pdfs cross: $1 - x = \frac{x}{4} \Rightarrow x = \frac{4}{5}$.

GATE 2013 | Q.50 | 2 marks

Common Data for Questions 50 and 51:
Consider the following figure.
Q.50 The current I_S in Amps in the voltage source, and voltage V_S in Volts across the current source respectively, are



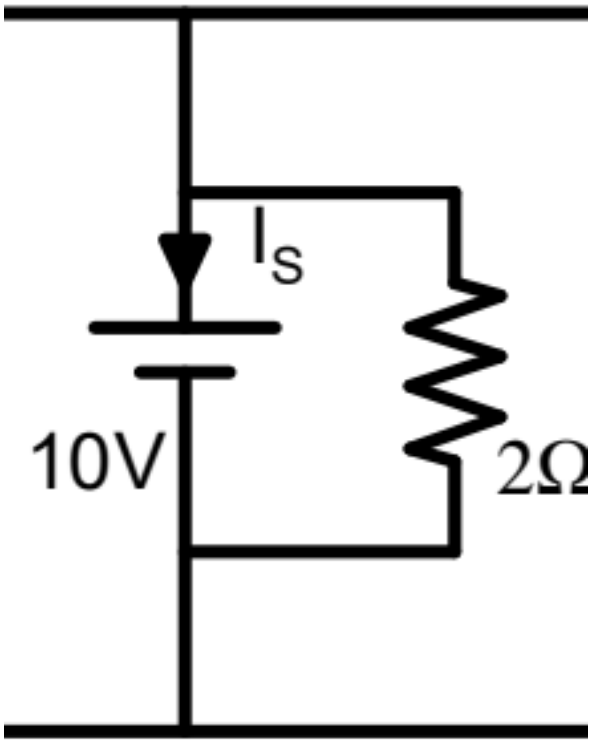
(A) 13, −20
(B) 8, −10
(C) −8, 20
(D) −13, 20

Answer: (D) [self-solved, no official key available]

Solution. The 10 V source sets the node at 10 V: the 2 Ω takes 5 A and the 1 Ω takes 10 A, while the current source supplies 2 A, so the voltage source supplies $I_S = -13$ A (as defined). The current source voltage is $V_S = 10 + 5 \times 2 = 20$ V.

GATE 2013 | Q.51 | 2 marks

Common Data for Questions 50 and 51 (figure above).
Q.51 The current in the 1 Ω resistor in Amps is



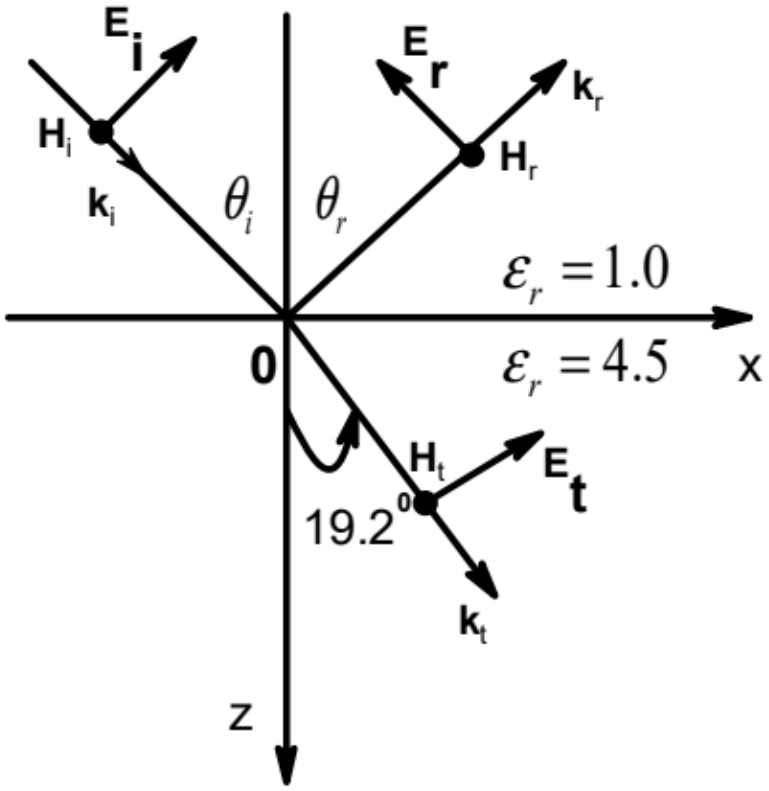
(A) 2
(B) 3.33
(C) 10
(D) 12

Answer: (C) [self-solved, no official key available]

Solution. The 1 Ω resistor is directly across the 10 V source: $I = 10/1 = 10$ A.

GATE 2013 | Q.52 | 2 marks

Statement for Linked Answer Questions 52 and 53:
A monochromatic plane wave of wavelength $\lambda = 600$ μm is propagating in the direction as shown in the figure below. $\vec{E}_i$, $\vec{E}_r$, and $\vec{E}_t$ denote incident, reflected, and transmitted electric field vectors associated with the wave.
Q.52 The angle of incidence θ_i and the expression for $\vec{E}_i$ are



(A) 60° and $\frac{E_0}{\sqrt{2}}(\hat{a}_x - \hat{a}_z)e^{-j\frac{\pi\times10^4(x+z)}{3\sqrt{2}}}$ V/m

(B) 45° and $\frac{E_0}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{-j\frac{\pi\times10^4 z}{3}}$ V/m

(C) 45° and $\frac{E_0}{\sqrt{2}}(\hat{a}_x - \hat{a}_z)e^{-j\frac{\pi\times10^4(x+z)}{3\sqrt{2}}}$ V/m

(D) 60° and $\frac{E_0}{\sqrt{2}}(\hat{a}_x - \hat{a}_z)e^{-j\frac{\pi\times10^4 z}{3}}$ V/m

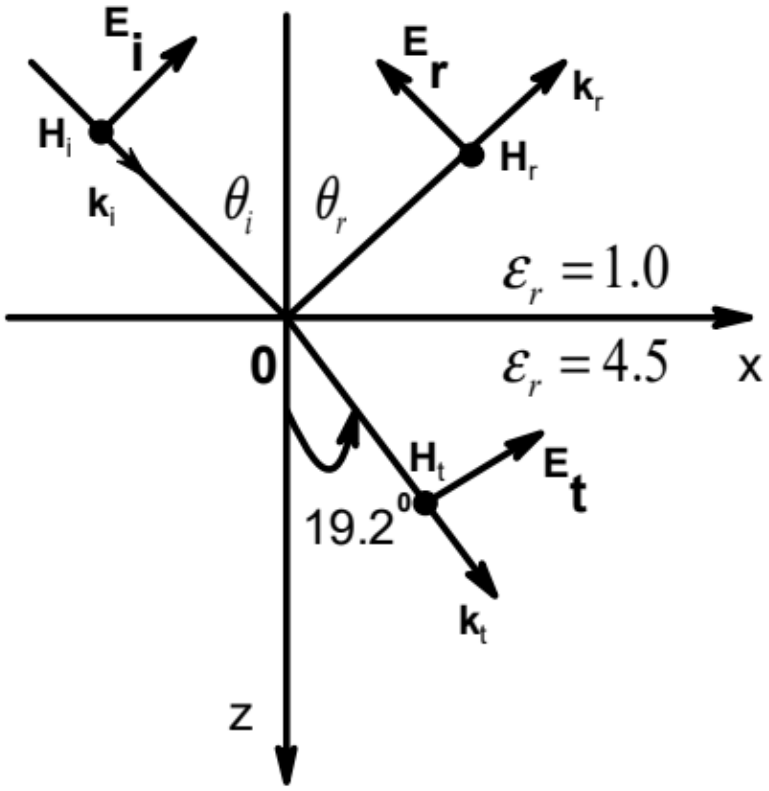
Answer: (C) [self-solved, no official key available]

Solution. Snell's law: $\sin\theta_i = \sqrt{4.5}\sin 19.2^\circ = 0.70 \Rightarrow \theta_i = 45^\circ$. With $k = \frac{2\pi}{\lambda} = \frac{\pi\times10^4}{3}$, the phase is $e^{-jk(x+z)/\sqrt{2}}$ and $\vec{E}_i \perp \vec{k}_i$ gives $(\hat{a}_x - \hat{a}_z)/\sqrt{2}$.

GATE 2013 | Q.53 | 2 marks

Statement for Linked Answer Questions 52 and 53 (figure above).

Q.53 The expression for $\vec{E}_r$ is



(A) $0.23\dfrac{E_0}{\sqrt{2}}(\hat{a}_x+\hat{a}_z)e^{-j\frac{\pi\times10^4(x-z)}{3\sqrt{2}}}$ V/m

(B) $-\dfrac{E_0}{\sqrt{2}}(\hat{a}_x+\hat{a}_z)e^{j\frac{\pi\times10^4 z}{3}}$ V/m

(C) $0.44\dfrac{E_0}{\sqrt{2}}(\hat{a}_x+\hat{a}_z)e^{-j\frac{\pi\times10^4(x-z)}{3\sqrt{2}}}$ V/m

(D) $\dfrac{E_0}{\sqrt{2}}(\hat{a}_x+\hat{a}_z)e^{-j\frac{\pi\times10^4(x+z)}{3}}$ V/m

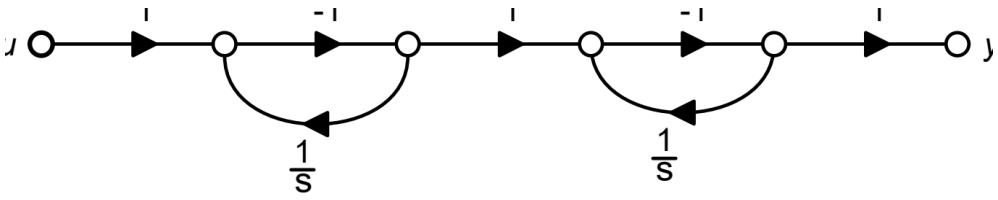
Answer: (A) [self-solved, no official key available]

Solution. For parallel polarisation $\Gamma=\dfrac{\eta_2\cos\theta_t-\eta_1\cos\theta_i}{\eta_2\cos\theta_t+\eta_1\cos\theta_i}$ with $\eta_2=\eta_0/\sqrt{4.5}$: $|\Gamma|=0.23$. The reflected wave travels in the $+x,-z$ direction.

GATE 2013 | Q.54 | 2 marks

Statement for Linked Answer Questions 54 and 55:
The state diagram of a system is shown below. A system is described by the state-variable equations $\dot{X}=AX+Bu;\ y=CX+Du$.

Q.54 The state-variable equations of the system shown in the figure above are



(A) $\dot{X}=\begin{bmatrix}-1 & 0\\ 1 & -1\end{bmatrix}X+\begin{bmatrix}-1\\ 1\end{bmatrix}u,\ y=[1\quad -1]X+u$

(B) $\dot{X} = \begin{bmatrix} -1 & 0 \\ -1 & -1 \end{bmatrix} X + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u,\ y = [-1 \quad -1]X + u$

(C) $\dot{X} = \begin{bmatrix} -1 & 0 \\ -1 & -1 \end{bmatrix} X + \begin{bmatrix} -1 \\ -1 \end{bmatrix} u,\ y = [-1 \quad 1]X - u$

(D) $\dot{X} = \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix} X + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u,\ y = [1 \quad -1]X - u$

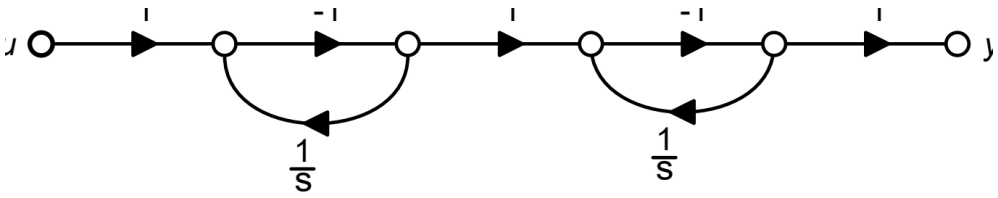
Answer: (A) [self-solved, no official key available]

Solution. Taking the integrator outputs as states: $\dot{x}_1 = -x_1 - u$, $\dot{x}_2 = x_1 - x_2 + u$, and $y = x_1 - x_2 + u$.

GATE 2013 | Q.55 | 2 marks

Statement for Linked Answer Questions 54 and 55 (figure above).

Q.55 The state transition matrix e^{At} of the system shown in the figure above is



(A) $\begin{bmatrix} e^{-t} & 0 \\ te^{-t} & e^{-t} \end{bmatrix}$

(B) $\begin{bmatrix} e^{-t} & 0 \\ -te^{-t} & e^{-t} \end{bmatrix}$

(C) $\begin{bmatrix} e^{-t} & 0 \\ e^{-t} & e^{-t} \end{bmatrix}$

(D) $\begin{bmatrix} e^{-t} & -te^{-t} \\ 0 & e^{-t} \end{bmatrix}$

Answer: (A) [self-solved, no official key available]

Solution. $A = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}$ is lower triangular with a repeated eigenvalue -1; $e^{At} = e^{-t}\begin{bmatrix} 1 & 0 \\ t & 1 \end{bmatrix}$.

GATE 2013 | Q.56 | 2 marks

Choose the grammatically CORRECT sentence:

(A) Two and two add four.
(B) Two and two become four.
(C) Two and two are four.
(D) Two and two make four.

Answer: (C) [self-solved, no official key available]

Solution. With a plural subject (two and two) the verb *are* is the grammatical choice.

GATE 2013 | Q.57 | 2 marks

Statement: You can always give me a ring whenever you need. Which one of the following is the best inference from the above statement?

(A) Because I have a nice caller tune.
(B) Because I have a better telephone facility.
(C) Because a friend in need is a friend indeed.

(D) Because you need not pay towards the telephone bills when you give me a ring.

Answer: (D) [self-solved, no official key available]

Solution. The offer to call whenever needed is best explained by the call being free of charge to the caller.

GATE 2013 | Q.58 | 2 marks

In the summer of 2012, in New Delhi, the mean temperature of Monday to Wednesday was 41°C and of Tuesday to Thursday was 43°C. If the temperature on Thursday was 15% higher than that of Monday, then the temperature in °C on Thursday was

(A) 40
(B) 43
(C) 46
(D) 49

Answer: (C) [self-solved, no official key available]

Solution. $M + T + W = 123$ and $T + W + Th = 129$ give $Th - M = 6$. With $Th = 1.15M$: $0.15M = 6 \Rightarrow M = 40$, $Th = 46$.

GATE 2013 | Q.59 | 2 marks

Complete the sentence: _____ Dare mistakes.

(A) commit
(B) to commit
(C) committed
(D) committing

Answer: (B) [self-solved, no official key available]

Solution. *Dare* is followed by the infinitive: *Dare to commit mistakes.*

GATE 2013 | Q.60 | 2 marks

They were requested not to quarrel with others. Which one of the following options is the closest in meaning to the word *quarrel*?

(A) make out
(B) call out
(C) dig out
(D) fall out

Answer: (D) [self-solved, no official key available]

Solution. *Quarrel* means to have a dispute: *fall out.*

GATE 2013 | Q.61 | 2 marks

A car travels 8 km in the first quarter of an hour, 6 km in the second quarter and 16 km in the third quarter. The average speed of the car in km per hour over the entire journey is

(A) 30
(B) 36
(C) 40
(D) 24

Answer: (C) [self-solved, no official key available]

Solution. Total distance = 30 km in 0.75 h: average speed = 40 km/h.

GATE 2013 | Q.62 | 2 marks

Find the sum to n terms of the series $10 + 84 + 734 + \ldots$

(A) $\frac{9}{9}(9^n + 1) + 1$

(B) $\frac{9}{8}(9^n - 1) + 1$

(C) $\frac{9}{8}(9^n - 1) + n$

(D) $\frac{9}{8}(9^n - 1) + n^2$

Answer: (D) [self-solved, no official key available]

Solution. The terms are $9^k + (2k - 1)$: $10 = 9 + 1$, $84 = 81 + 3$, $734 = 729 + 5$. The sum is $\frac{9}{8}(9^n - 1) + n^2$.

GATE 2013 | Q.63 | 2 marks

Statement: There were different streams of freedom movements in colonial India carried out by the moderates, liberals, radicals, socialists, and so on. Which one of the following is the best inference from the above statement?

(A) The emergence of nationalism in colonial India led to our Independence.

(B) Nationalism in India emerged in the context of colonialism.

(C) Nationalism in India is homogeneous.

(D) Nationalism in India is heterogeneous.

Answer: (D) [self-solved, no official key available]

Solution. Several distinct movements imply that Indian nationalism was not uniform, i.e. heterogeneous.

GATE 2013 | Q.64 | 2 marks

The set of values of p for which the roots of the equation $3x^2 + 2x + p(p - 1) = 0$ are of opposite sign is

(A) $(-\infty, 0)$

(B) $(0, 1)$

(C) $(1, \infty)$

(D) $(0, \infty)$

Answer: (B) [self-solved, no official key available]

Solution. The roots have opposite signs iff their product $\frac{p(p-1)}{3} < 0$, i.e. $0 < p < 1$.

GATE 2013 | Q.65 | 2 marks

What is the chance that a leap year, selected at random, will contain 53 Saturdays?

(A) 2/7

(B) 3/7

(C) 1/7

(D) 5/7

Answer: (A) [self-solved, no official key available]

Solution. A leap year has 52 weeks + 2 extra days; one of them is a Saturday in 2 of the 7 equally likely cases: 2/7.